PigeOff

Technical Development

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Process so far

- Opted for **Radar** & **Cam**
  - Mic too unreliable
  - LoRa to map city-wide pigeon activity
  - Expected high false positive rate
- Currently working on:
  - Setting up camera
  - Setting up classifier
  - Setting up LoRa communication
  - Measuring!

Measuring

- Average amount of pigeons an hour/day
- Energy consumption in different states
  a. Idle (radar on)
  b. Taking a picture
  c. Transmitting results

Component Evaluation

- **ESP32-CAM (OV2640)**
  - Picking a resolution (96, 160)
  - ~200 - 250 mA active
- **HLK-LD2410S (Radar)**
  - ~0.1 mA
- **Heltec Board**
  - ~0.8 - 1.0 mA (sleep)
  - Deciding on what to send
  - ~120 - 150 mA (LoRa)

If 12 triggers an hour: ~33 Days

Demo